

Summer 1980

Problem 1 (Su80). Exhibit a real 3×3 matrix having minimal polynomial $(t^2 + 1)(t - 10)$, which, as a linear transformation of $\mathbb{R}^3$, leaves invariant the line L through $(0, 0, 0)$ and $(1, 1, 1)$ and the plane through $(0, 0, 0)$ perpendicular to L .

Problem 2 (Su80). Which of the following matrix equations have a real matrix solution X ? (It is not necessary to exhibit solutions.)

$$X^3 = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 2 & 3 & 0 \end{pmatrix} \quad 2X^5 + X = \begin{pmatrix} 3 & 5 & 0 \\ 5 & 1 & 9 \\ 0 & 9 & 0 \end{pmatrix}$$

$$X^6 + 2X^4 + 10X = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad X^4 = \begin{pmatrix} 3 & 4 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -3 \end{pmatrix}$$

Problem 3 (Su80). Let $T : V \rightarrow V$ be an invertible linear transformation of a vector space V . Denote by G the group of all maps $f_{k,a} : V \rightarrow V$ where $k \in \mathbb{Z}$, $a \in V$, and for $x \in V$,

$$f_{k,a}(x) = T^k x + a \quad x \in V$$

Prove that the commutator subgroup G' of G is isomorphic to the additive group of the vector space $(T - I)V$, the image of $T - I$. G' is generated by all $ghg^{-1}h^{-1}$, g and h in G .

Problem 4 (Su80, Sp12, Sp21, Sp22, Fa22). Let G be a finite group and H be a subgroup.

1. Show that the number of subgroups of G conjugate to H divides the index of H .
2. Show that if $G = \bigcup_{g \in G} gHg^{-1}$ then $G = H$.

Problem 5 (Su80). Consider the differential equations

$$\frac{dx}{dt} = -x + y \quad \frac{dy}{dt} = \log(20 + x) - y$$

Let $x(t)$ and $y(t)$ be a solution defined for all $t \geq 0$ with $x(0) > 0$ and $y(0) > 0$. Prove that $x(t)$ and $y(t)$ are bounded.

Problem 6 (Su80). Let C denote the positively oriented circle $|z| = 2$, $z \in \mathbb{C}$. Evaluate the integral

$$\int_C \sqrt{z^2 - 1} \, dz$$

where the branch of the square root is chosen so that $\sqrt{2^2 - 1} > 0$.

Problem 7 (Su80). Exhibit a conformal map from $\{z \in \mathbb{C} \mid |z| < 1, \Re z > 0\}$ onto $\mathbb{D} = \{z \in \mathbb{C} \mid |z| < 1\}$.

Problem 8 (Su80). Give an example of a subset of $\mathbb{R}$ having uncountably many connected components. Can such a subset be open? Closed?

Problem 9 (Su80, Sp97, Fa11, Fa13, Fa17, Sp21, Fa25). For each $(a, b, c) \in \mathbb{R}^3$, consider the series

$$\sum_{n=3}^{\infty} \frac{a^n}{n^b \log^c n}$$

Determine the values of (a, b, c) for which the series converges absolutely; converges but not absolutely; diverges.

Problem 10 (Su80). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a function whose partial derivatives of order ≤ 2 are everywhere defined and continuous.

1. Let $a \in \mathbb{R}^n$ be a critical point of f , i.e., $\partial f / \partial x_j(a) = 0$, $i = 1, \dots, n$. Prove that a is a local minimum provided the Hessian matrix

$$\left(\frac{\partial^2 f}{\partial x_i \partial x_j} \right)$$

is positive definite at $x = a$.

2. Assume the Hessian matrix is positive definite at all x . Prove that f has, at most, one critical point.

Problem 11 (Su80, Fa96, Sp11). Prove that every finite group is isomorphic to

1. A subgroup of a group of permutations of n objects;
2. A subgroup of the group of even permutations of n objects.

Problem 12 (Su80). Let $S \subset \mathbb{R}^3$ denote the ellipsoidal surface defined by

$$2x^2 + (y - 1)^2 + (z - 10)^2 = 1$$

Let $T \subset \mathbb{R}^3$ be the surface defined by

$$z = \frac{1}{x^2 + y^2 + 1}$$

Prove that there exist points $p \in S$, $q \in T$, such that the line $\overline{pq}$ is perpendicular to S at p and to T at q .

Problem 13 (Fa79, Su80, Fa82). Consider the polynomial ring $\mathbb{Z}[x]$ and the ideal $\mathfrak{J}$ generated by 7 and $x - 3$.

1. Show that for each $r \in \mathbb{Z}[x]$, there is an integer α satisfying $0 \leq \alpha \leq 6$ such that $r - \alpha \in \mathfrak{J}$.
2. Find α in the special case $r = x^{250} + 15x^{14} + x^2 + 5$.

Problem 14 (Su80, Fa96). Let A and B be real 2×2 matrices such that $A^2 = B^2 = I$ and $AB + BA = 0$. Prove there exists a real nonsingular matrix T with

$$TAT^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad TBT^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Problem 15 (Su80, Fa92). Let E be a finite dimensional vector space over a field $\mathbb{F}$. Suppose $B : E \times E \rightarrow \mathbb{F}$ is a bilinear map, not necessarily symmetric. Define subspaces

$$E_1 = \{x \in E \mid B(x, y) = 0 \text{ for all } y \in E\}$$

$$E_2 = \{y \in E \mid B(x, y) = 0 \text{ for all } x \in E\}$$

Prove that $\dim E_1 = \dim E_2$.

Problem 16 (Su80). Let (a_n) be a sequence of nonzero real numbers. Prove that the sequence of functions $f_n : \mathbb{R} \rightarrow \mathbb{R}$

$$f_n(x) = \frac{1}{a_n} \sin(a_n x) + \cos(x + a_n)$$

has a subsequence converging to a continuous function.

Problem 17 (Su80). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be monotonically increasing (perhaps discontinuous). Suppose $0 < f(0)$ and $f(100) < 100$. Prove $f(x) = x$ for some x .

Problem 18 (Su80, Fa83, Sp84, Fa87, Fa96, Fa07). How many zeros does the complex polynomial

$$3z^9 + 8z^6 + z^5 + 2z^3 + 1$$

have in the annulus $1 < |z| < 2$?

Problem 19 (Su80, Fa17, Fa21, Fa24). Let f be a meromorphic function on $\mathbb{C}$ which is analytic in a neighborhood of 0. Let its Maclaurin series be

$$\sum_{k=0}^{\infty} a_k z^k$$

with all $a_k \geq 0$. Suppose there is a pole of modulus $r > 0$ and no pole has modulus $< r$. Prove there is a pole at $z = r$.

Problem 20 (Su77, Su80, Sp82, Sp83). Prove that the initial value problem

$$\frac{dx}{dt} = 3x + 85 \cos x \quad x(0) = 77$$

has a solution $x(t)$ defined for all $t \in \mathbb{R}$.